

Lab 6: Map

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Each 2D thread transforms one grayscale pixel: threshold 128 for binarization, clamped $1.2p + 30$ for brightness, or $0.5I_1 + 0.5I_2$ for blending. The second image is resized to match the first. Several 2D blocks are timed, and NumPy references validate the saved outputs.

Kernel timing uses `time.time()` after JIT warm-up and CUDA synchronization, averaged over three GPU runs. Device allocation and host/device copies are excluded. CPU measurements are single runs. Speedup is relative to the stated CPU implementation. Actual measurements appear after running the notebook on Colab; no timings are assumed here.

block	operation	gpu_ms	max_error
(8, 8)	binarized	0.062863	0
(8, 8)	brightness	0.0594457	0
(8, 8)	blended	0.093619	0
(16, 16)	binarized	0.0465711	0
(16, 16)	brightness	0.05428	0
(16, 16)	blended	0.0584126	0
(32, 16)	binarized	0.0404517	0
(32, 16)	brightness	0.0443459	0
(32, 16)	blended	0.0549952	0
(32, 32)	binarized	0.0649293	0
(32, 32)	brightness	0.0596841	0
(32, 32)	blended	0.0685056	0

